

BIOLOGY — GRADE 10 | PLANT NUTRITION

Leaf Structure

The architecture of a photosynthetic machine

Today we find the first real evidence for why our sukuma wiki wilts.

What's actually inside a leaf?

- Take a fresh mango or bean leaf and a hand lens. Look at both surfaces. Feel them.
- Without cutting it, sketch what you think a cross-section (a slice through the middle) would look like.
- Label any layers you think are there, and write one sentence on why you think they are there.

Compare with your partner: one similarity, one difference. We come back to your sketch at the end.

A leaf must let light in — without drying out.

Cuticle

A waxy, transparent, waterproof layer on the outer surface. Light passes through it, but it slows water loss from the leaf.

Epidermis

A single layer of tightly packed, transparent cells under the cuticle. It protects the leaf. It has almost no chloroplasts, so light passes straight through to the cells below.

Activity: with your hand lens, find the shinier side of the leaf. That is the upper surface — thicker cuticle, facing the sun.

QUICK CHECK 1

Why is it useful that the cuticle is both waxy and transparent?

A It lets light in while reducing water loss

B It traps carbon dioxide inside the leaf

C It makes food for the leaf

D It carries water to the leaf cells

Answer key is teacher-side only, in a separate file.

Two layers of cells, two different jobs.

Palisade mesophyll

Tall cells packed tightly just under the upper epidermis, full of chloroplasts. Most photosynthesis happens here: tight packing puts the most chloroplasts where the light is strongest.

Spongy mesophyll

Loosely packed, irregular cells with large air spaces between them. The air spaces let carbon dioxide and oxygen move easily to and from every cell.

Microscope (100×): sketch one palisade cell and one spongy cell. Which has more green chloroplasts?

QUICK CHECK 2

Which layer is packed tightly with chloroplasts to capture the most light?

- A** Lower epidermis
- B** Palisade mesophyll
- C** Spongy mesophyll
- D** Cuticle

Answer key is teacher-side only, in a separate file.

Stomata: tiny pores that do two jobs at once.

- Stomata are tiny pores, found mostly in the lower epidermis. Look for them under the microscope.
- Each stoma sits between two guard cells, which control how wide the pore opens.
- When a stoma is open, carbon dioxide enters for photosynthesis and oxygen leaves.
- But an open pore works both ways: water vapour also escapes through it. This water loss is called transpiration.

QUICK CHECK 3

What controls how wide a stoma opens?

- A** The cuticle
- B** The two guard cells around it
- C** The palisade cells
- D** The xylem vessels

Answer key is teacher-side only, in a separate file.

Vascular bundles: the leaf's delivery pipes.

Xylem

Carries water (and dissolved minerals) up from the roots into the leaf, where every cell needs it.

Phloem

Carries the food made in the leaf (sugars) away to the rest of the plant.

The veins you can see on the leaf are vascular bundles. Find one in your cross-section.

QUICK CHECK 4

Which tissue carries water from the roots up into the leaf?

A Phloem

B Xylem

C Palisade mesophyll

D Epidermis

Answer key is teacher-side only, in a separate file.

So why does the sukuma wiki wilt at midday?

1. Xylem keeps delivering water from the roots to the leaf cells.
2. In bright midday light, stomata open wide for gas exchange, so water vapour escapes faster.
3. When water leaves faster than xylem can replace it, the cells lose water and go soft — the leaf wilts.
4. At night stomata close, water loss slows, the xylem catches up, and the plant recovers by morning.

Jigsaw: your group explains ONE structure in 60 seconds — what it looks like, what it does, and how it links to wilting.

QUICK CHECK 5

Using today's evidence, why does the sukuma wiki wilt at midday even though the soil is moist?

- A** The soil dries out faster in strong sunlight
- B** Water vapour leaves through open stomata faster than xylem can replace it
- C** The roots stop absorbing water when it is hot
- D** Chlorophyll breaks down in direct sunlight

Answer key is teacher-side only, in a separate file.

Update your model. Add one new question.

UPDATE YOUR MODEL

Add a leaf cross-section to your Lesson 1 model: cuticle, epidermis, palisade, spongy mesophyll, stoma with guard cells, and a vascular bundle. Draw arrows for light in, water up the xylem, and water vapour out of the stoma. Finish: “My model still cannot explain ____” and circle it in red.

DRIVING QUESTION BOARD

Yellow note: one NEW question today raised (e.g. What makes guard cells open or close?).

Green note: one question from Lesson 1 you can now partly answer — write the partial answer.

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TED-Ed • Unclear match: this generic title appears in 20 lessons across the corpus. Preview before using.



READING [Seasonal Changes in Plants: Leaf Pigment \(PLIX\)](#)

CK-12 • Partial match: about leaf pigments, not leaf structure



SEARCH [Search ARES: leaf structure, stomata](#)

Best option for this lesson until the link fix lands

Links open only on the school network (ares.local). Click to open during the lesson.

5 questions asked during class today

1. Why is the cuticle both waxy and transparent?
2. Which layer is packed with chloroplasts to capture the most light?
3. What controls how wide a stoma opens?
4. Which tissue carries water up into the leaf?
5. Why does the sukuma wiki wilt at midday even though the soil is moist?

Ask your teacher how to submit your answers today (for example, on a paper answer slip).